Enterprise Network Infrastructure

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Date: February 2026

Technologies:

- IPv4 (VLSM)
- HSRP - First Hop Redundancy (per-VLAN load balancing)
- EtherChannel LACP - Link aggregation
- Rapid PVST+ - Spanning Tree with PortFast & BPDU Guard
- Inter-VLAN Routing - SVI & Router on a stick (RoaS)
- Port Security + DHCP snooping - Layer 2 Security
- DHCP Server - Centralized Address Assignment

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1. Introduction and Objectives

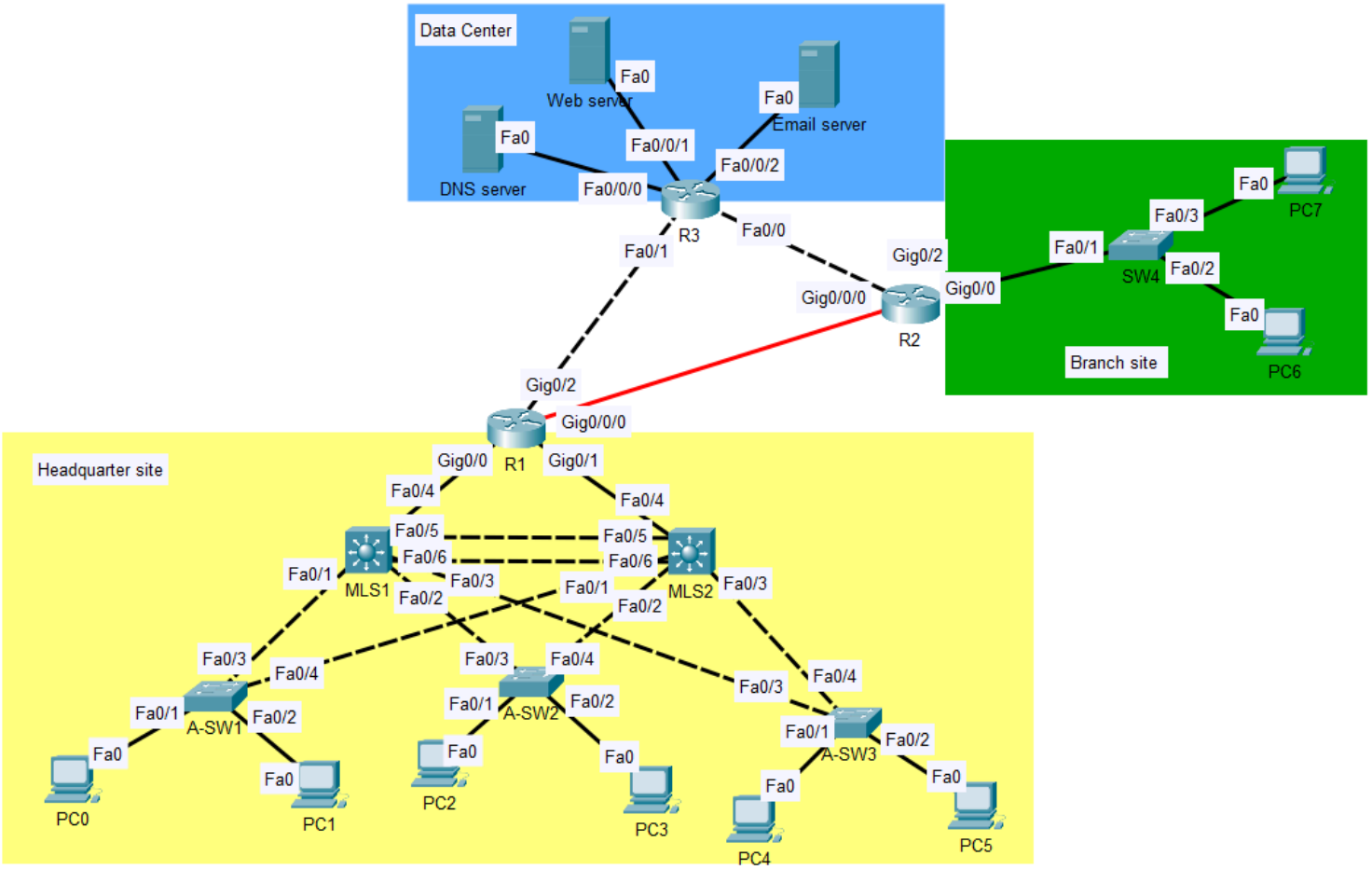
This project demonstrates the design and implementation of a scalable, high-availability enterprise network infrastructure. The architecture is divided into three interconnected sites, each serving a specific functional role: a redundant Core/Distribution block, a Branch office using Router-on-a-Stick, and a dedicated Data Center for network services.

The primary objective was to ensure 24/7 operational continuity by implementing Layer 2 and Layer 3 redundancy protocols such as EtherChannel, Spanning Tree (STP), and HSRP. Furthermore, the network is built on an IPv4 foundation to meet modern connectivity standards, featuring robust security measures including Port Security and DHCP Snooping to protect the internal infrastructure from common vulnerabilities.

At the end of this project, the following objectives were achieved:

- Designed an IPv4 IP addressing plan using VLSM.
- Configured VLANs, trunking, and EtherChannel across the Headquarter switch block.
- Implemented Inter-VLAN Routing using SVI on MLS switches (HQ) and Router-on-a-Stick (Branch Office).
- Configured HSRP for first-hop redundancy with per-VLAN load balancing.
- Managed dynamic address allocation using DHCPv4
- Configured SSH remote management on MLS1 and MLS2.
- Applied Layer 2 security mechanisms: Port Security, DHCP Snooping, BPDU Guard, and DTP disable.
- Configured and tested DNS, HTTP, HTTPS, and EMAIL services on Data Center servers.
- Verified end-to-end connectivity across all three sites.

2. Topology and Network diagram



2.1 Network Connectivity

The interconnection between sites is made using three routers: **R1** for the Headquarters, **R2** for the Branch Site, and **R3** for the Data Center.

- **Data Center Connection (R3):** For the Data Center, we used a **Cisco 1841** router. Because this model only has 2 FastEthernet interfaces, it limited our physical connections. To solve this, we added an **HWIC-4ESW** module, which provides 4 switching ports. This allowed us to connect the Servers directly to the router without needing an extra switch, making the best use of our available equipment.
- **HQ to Branch Connection (R1-R2):** Between the Headquarters (R1) and the Branch Office (R2), we used **HWIC-1C-SFP** modules. We chose these modules to establish a fiber optic connection, ensuring a faster link speed between these two main sites for better internal communication.

2.2 Network devices list

Note: For the purpose of this project, only 6 PCs were used to keep the topology clear and simple. In a real-world environment, each VLAN would naturally include a much larger number of end-devices.

Location (Site)	Host Name	Equipment model	Topology role	Key functions
Site 1 (HQ)	**R1**	Cisco 2911	Core Router	Inter-site routing
	MLS1,2	Cisco 3560	Distribution Switch	HSRP Active/Standby, Inter-VLAN Routing.
	A-SW1/2/3	Cisco 2960	Access Switch	User access, VLANs.
Site 2 (BO)	**R2**	Cisco 2911	Branch Router	Branch Office gateway, RoaS.
	SW4	Cisco 2960	Access Switch	Extend Branch Office network.
Site 3 (DC)	**R3**	Cisco 2911	DC Router	Servers aggregation, Routing services.
	DNS/Web/Mail	Generic Server	Application Servers	Hosting critical services (HTTP, DNS).

3. VLAN Addressing Plan

Each site is segmented into multiple VLANs for broadcast domain reduction, logical organization, and security. Two special VLANs are defined across all sites: VLAN 666 (UnusedPorts) to park all inactive ports, and VLAN 555 (Native-BH) as a blackhole native VLAN to prevent VLAN hopping attacks.

Site 1 - Headquarter

VLAN ID	VLAN name	IPv4 Network address
10	IT	10.1.10.0 /24
20	HR	10.1.20.0 /24
30	Sales	10.1.30.0 /24
99	MGMT	10.1.99.0 /29
666	UnusedPorts	-
555	Native-BH	-

Site 2 - Branch Office

VLAN ID	VLAN name	Network address
110	Support	10.2.110.0/24
120	Sales	10.2.120.0/24
666	UnusedPorts	-
555	Native-BH	-

Site 3 - Data Center - IPv4

VLAN ID	VLAN name	Network address
200	DNS_Server	10.3.200.0/24
210	Web_Server	10.3.210.0/24
220	Email_server	10.3.220.0/24

4.IP addressing table

Standardized Gateway Allocation: *To ensure a consistent and intuitive addressing scheme, the first assignable IP address in every subnet has been reserved as the Default Gateway. This standardization simplifies network management and troubleshooting.*

Redundancy and Virtual IP (VIP): *It should be noted that many end-devices use **10.1.x.1** (where x is VLAN id) as their Default Gateway. Although this specific address is not assigned to a single physical device in the addressing table, it functions as a **Virtual IP (VIP)**. This is implemented via the **HSRP (Hot Standby Router Protocol)** to provide first-hop redundancy and ensure network availability in the event of a hardware failure*

4.1 IPv4 addressing

Subnetting Rationale and Strategy

The IPv4 addressing plan was designed using **VLSM (Variable Length Subnet Masking)** to balance scalability with efficient address management. The following logic was applied:

- **User VLANs & Data Center (/24 - 255.255.255.0):** A **/24 mask** was chosen for all user-facing departments (IT, HR, Sales) and the Data Center. While the current number of devices is small, this prefix provides up to 254 usable addresses per subnet. This strategy ensures **future scalability**, allowing the company to add more workstations or servers without the need for disruptive network re-addressing or downtime. It also simplifies the routing table and management.
- **Management VLAN (/28 - 255.255.255.240):** For the Management network, a **/28 prefix** was implemented to strike an ideal balance between security and **future scalability**. While a more restrictive mask like /29 provides a smaller attack surface, it offers only 6 usable addresses, which would limit the ability to expand the infrastructure.
- **WAN Transport Links (/30 - 255.255.255.252):** Point-to-point connections between routers use a **/30 mask**. This is the most efficient way to interconnect two routing nodes, as it provides exactly 2 usable IP addresses, eliminating any waste of the IP address space.

Site 1 - Headquarter

Device	Interface	IP address	Subnet mask	Gateway
R1	Gig0/0	10.1.0.1	255.255.255.252	-
R1	Gig0/1	10.1.0.5	255.255.255.252	-
MLS1	Fa0/4	10.1.0.2	255.255.255.252	10.1.0.1
MLS1	Vlan 10	10.1.10.2	255.255.255.0	10.1.0.1
MLS1	Vlan 20	10.1.20.2	255.255.255.0	10.1.0.1
MLS1	Vlan 30	10.1.30.2	255.255.255.0	10.1.0.1
MLS1	Vlan 99	10.1.99.2	255.255.255.240	-
MLS2	Fa0/4	10.1.0.6	255.255.255.252	10.1.0.5
MLS2	Vlan 10	10.1.10.3	255.255.255.0	10.1.0.5
MLS2	Vlan 20	10.1.20.3	255.255.255.0	10.1.0.5
MLS2	Vlan 30	10.1.30.3	255.255.255.0	10.1.0.5
MLS2	Vlan 99	10.1.99.3	255.255.255.240	-
A-SW1	Vlan 99	10.1.99.4	255.255.255.240	10.1.99.1 (VIP)
A-SW2	Vlan 99	10.1.99.5	255.255.255.240	10.1.99.1 (VIP)
A-SW3	Vlan 99	10.1.99.6	255.255.255.240	10.1.99.1 (VIP)
PC1	NIC	DHCP assigned	255.255.255.0	10.1.10.1
PC2	NIC	DHCP assigned	255.255.255.0	10.1.10.1
PC3	NIC	DHCP assigned	255.255.255.0	10.1.20.1
PC4	NIC	DHCP assigned	255.255.255.0	10.1.20.1
PC5	NIC	DHCP assigned	255.255.255.0	10.1.30.1
PC6	NIC	DHCP assigned	255.255.255.0	10.1.30.1

Site 2 - Branch Office

Device	Interface	IP address	Subnet mask	gateway
R2	Gig0/0.110	10.2.110.1	255.255.255.0	-
R2	Gig0/0.120	10.2.120.1	255.255.255.0	-
PC6	NIC	10.2.110.10	255.255.255.0	10.2.110.1
PC7	NIC	10.2.120.10	255.255.255.0	10.2.120.1

Site 3 - Data Center

Device	Interface	IP address	Subnet mask	Gateway
R3	Vlan 200	10.3.200.1	255.255.255.248	-
R3	Vlan 210	10.3.210.1	255.255.255.248	-
R3	Vlan 220	10.3.220.1	255.255.255.248	-
DNS server	NIC	10.3.200.10	255.255.255.248	10.3.200.1
Web server	NIC	10.3.210.10	255.255.255.248	10.3.210.1
Email server	NIC	10.3.220.10	255.255.255.248	10.3.220.1

WAN connections

Device Name	Interface	IPv4 address/CIDR	Subnet mask	Gateway
R1	Gig0/0/0	10.0.0.9/30	255.255.255.252	10.0.0.9
R1	Gig0/2	10.0.0.13/30	255.255.255.252	10.0.0.13
R2	Gig0/0/0	10.0.0.10/30	255.255.255.252	10.0.0.9
R2	Gig0/2	10.0.0.17/30	255.255.255.252	10.0.0.17
R3	Fa0/0	10.0.0.18/30	255.255.255.252	10.0.0.17
R3	Fa0/1	10.0.0.14/30	255.255.255.252	10.0.0.13

5. Core configurations

5.1. EtherChannel (Link Aggregation)

To eliminate bottlenecks and provide link redundancy, LACP (Link Aggregation Control Protocol) is implemented on critical backbone links. This ensures that if one physical cable fails, the logical connection remains active without interrupting the Spanning Tree topology.

Site 1 (HQ) is the only site where Etherchannel is implemented (MLS1 and MLS2). Below the appropriate configuration is shown:

Representative device: MLS1 (same configuration applies to MLS2)

```
MLS1(config)#in range f0/5 - 6
MLS1(config-if)#channel-group 1 mode active
```

To verify EtherChannel configuration we use ‘show etherchannel summary’ command on one of the Layer 3 switches, and this should be the expected output:

```
MLS2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Fa0/5(P) Fa0/6(P)
```

5.2 VLAN & Trunking implementation

To ensure an efficient and organized network environment, each site in the Enterprise Network infrastructure is segmented into multiple Virtual LANs (VLANs). This logical separation is fundamental for:

- ***Broadcast Domain Reduction:*** Limiting broadcast traffic to its respective department, which prevents network congestion and improves overall performance.

- ***Logical Organization:*** Aligning the network topology with the physical and functional structure of the company (IT, HR, Sales, and Management).

The following design principles were applied:

• ***Encapsulation:*** Industry-standard IEEE 802.1Q is used for all trunk links to ensure efficient frame tagging between switches.

• ***Security (Native VLAN):*** In Headquarter, VLAN 555 has been designated as the Native VLAN for all trunk ports. By using a non-default, unused VLAN for untagged traffic, the network is hardened against VLAN Hopping attempts.

• ***Trunk Pruning:*** Only the necessary VLANs (e.g 10, 20, 30, 99 for HQ) are permitted on the uplinks to prevent unnecessary traffic from leaking into other segments of the infrastructure.

5.2.1 Headquarter (HQ) Configuration

Representative device: A-SW1 (access layer) — A-SW2 and A-SW3 are configured identically, with different port-to-VLAN assignments

```
A-SW1(config)# vlan 10
A-SW1(config-vlan)# name IT

A-SW1(config-vlan)# vlan 20
A-SW1(config-vlan)# name HR

A-SW1(config-vlan)# vlan 30
A-SW1(config-vlan)# name Sales

A-SW1(config-vlan)# vlan 99
A-SW1(config-vlan)# name MGMT

A-SW1(config-vlan)# vlan 666
A-SW1(config-vlan)# name UnusedPorts

A-SW1(config-vlan)# vlan 555
A-SW1(config-vlan)# name Native-BH
A-SW1(config-vlan)# exit


A-SW1(config)# in range f0/5 - 24, g0/1 - 2
A-SW1(config-if)# switchport mode acceess
A-SW1(config-if)# switchport access vlan 666
A-SW1(config-if)# shut
A-SW1(config-if)# exit

A-SW1(config)# in range f0/1 - 2
A-SW1(config-if)# switchport mode access
A-SW1(config-if)# switchport access vlan 10
A-SW1(config-if)# exit

A-SW1(config)# in range f0/3 - 4
A-SW1(config-if)# switchport mode trunk
A-SW1(config-if)# switchport trunk allowed vlan 10,20,30,99,555
A-SW1(config-if)# switchport trunk native vlan 555
A-SW1(config-if)# exit
```

Below we have the verification and the expected output using proper show commands

```
A-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   IT                               active    Fa0/1, Fa0/2
20   HR                               active
30   Sales                            active
99   MGMT                             active
555  Native-BH                        active
666  UnusedPorts                      active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
A-SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/3       on           802.1q         trunking      555
Fa0/4       on           802.1q         trunking      555

Port        Vlans allowed on trunk
Fa0/3       10,20,30,99,555
Fa0/4       10,20,30,99,555

Port        Vlans allowed and active in management domain
Fa0/3       10,20,30,99,555
Fa0/4       10,20,30,99,555

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/3       20,555
Fa0/4       10,30,99
```

MLS1 configuration is shown as a representative configuration for the HQ distribution layer (MLS2 is configured identically):

```
MLS1(config)# vlan 10
MLS1(config-vlan)# name IT

MLS1(config-vlan)# vlan 20
MLS1(config-vlan)# name HR

MLS1(config-vlan)# vlan 30
MLS1(config-vlan)# name Sales

MLS1(config-vlan)# vlan 99
MLS1(config-vlan)# name MGMT

MLS1(config-vlan)# vlan 666
MLS1(config-vlan)# name UnusedPorts

MLS1(config-vlan)# vlan 555
MLS1(config-vlan)# name Native-BH
MLS1(config-vlan)# exit


MLS1(config)# in range f0/7 - 24, g0/1 - 2
MLS1(config-if)# switchport mode access
MLS1(config-if)# switchport access vlan 666
MLS1(config-if)# shut

MLS1(config)# in range f0/1 - 3
MLS1(config-if-range)# switchport trunk encapsulation dot1q
MLS1(config-if-range)# switchport mode trunk
MLS1(config-if-range)# switchport trunk allowed vlan 10,20,30,99,555
MLS1(config-if-range)# switchport trunk native vlan 555
MLS1(config-if-range)# exit

MLS1(config)# in range f0/5 - 6
MLS1(config-if-range)# channel-group 1 mode active
MLS1(config-if-range)# exit

MLS1(config)# in po1
MLS1(config-if)# switchport trunk encapsulation dot1q
MLS1(config-if)# switchport mode trunk
MLS1(config-if)# switchport trunk allowed vlan 10,20,30,99,555
MLS1(config-if)# switchport trunk native vlan 555
MLS1(config-if)# no shut
MLS1(config-if)# exit
```

Same commands as for A-SW1 are used to verify the configuration and the expected output should be:

```
MLS1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   IT                               active
20   HR                               active
30   Sales                            active
99   MGMT                             active
555  Native-BH                        active
666  UnusedPorts                      active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
MLS1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      555
Fa0/1       on           802.1q         trunking      555
Fa0/2       on           802.1q         trunking      555
Fa0/3       on           802.1q         trunking      555

Port        Vlans allowed on trunk
Po1         10,20,30,99,555
Fa0/1       10,20,30,99,555
Fa0/2       10,20,30,99,555
Fa0/3       10,20,30,99,555

Port        Vlans allowed and active in management domain
Po1         10,20,30,99,555
Fa0/1       10,20,30,99,555
Fa0/2       10,20,30,99,555
Fa0/3       10,20,30,99,555

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,99,555
Fa0/1       10,20,30,99,555
Fa0/2       10,20,30,99,555
Fa0/3       10,20,30,99,555
```

5.2.2 Branch Office Configuration

For the Branch Office, a single-switch access model is implemented. The configuration ensures that even in a smaller site, the enterprise security policies—such as **VLAN Parking** and **Native VLAN isolation**—are strictly maintained.

Representative Device: SW4

- **VLAN 110 & 120:** Assigned to local Support and Sales departments.
- **Uplink (f0/1):** Configured as a trunk link to the Branch Router, allowing only necessary traffic.

```
SW4(config)# vlan 110
SW4(config-vlan)# name Support
SW4(config-vlan)# vlan 120
SW4(config-vlan)# name Sales
SW4(config-vlan)# vlan 666
SW4(config-vlan)# name UnusedPorts
SW4(config-vlan)# vlan 555
SW4(config-vlan)# name Native-BH
SW4(config-vlan)# exit

SW4(config)# in range f0/4 - 24, g0/1 - 2
SW4(config-if)# switchport mode access
SW4(config-if)# switchport access vlan 666
SW4(config-if)# shut

SW4(config)# in f0/2
SW4(config-if)# switchport mode access
SW4(config-if)# switchport access vlan 110

SW4(config)# in f0/3
SW4(config-if)# switchport mode access
SW4(config-if)# switchport access vlan 120

SW4(config)# in f0/1
SW4(config-if)# switchport mode trunk
SW4(config-if)# switchport trunk allowed vlan 110,120,555
SW4(config-if)# switchport trunk native vlan 555
```

note: same verification commands are used but for the sake of space their output will not be included (same applies for 5.2.3)

5.2.3 Data Center Configuration

Representative device: R3 (HWIC-4ESW module)

```
R3 (vlan)# vlan 200 name DNS_Server
VLAN 200 modified:
    Name: DNS_Server

R3 (vlan)# vlan 210 name Web_Server
VLAN 210 modified:
    Name: DNS_Server

R3 (vlan)# vlan 220 name Email_Server
VLAN 220 modified:
    Name: DNS_Server
R3 (vlan)# exit


R3# configure terminal
R3(config)# interface f0/0/0
R3(config-if)# switchport mode access
R3(config-if)# switchport access vlan 200
R3(config-if)# exit


R3(config)# in f0/0/1
R3(config-if)# switchport mode access
R3(config-if)# switchport access vlan 210
R3(config-if)# exit

R3(config)# in f0/0/2
R3(config-if)# switchport mode access
R3(config-if)# switchport access vlan 220
R3(config-if)# exit

R3(config)# in f0/0/3
R3(config-if)# shutdown
R3(config-if)# switchport mode acess
R3(config-if)# switchport access vlan 666
R3(config-if)# exit
```

5.3 STP and portfast Configuration

Rapid PVST+ is used to prevent Layer 2 loops by electing a Root Bridge per VLAN based on Bridge Priority. To implement per-VLAN load balancing, MLS1 is configured as Root Bridge for VLAN 10 and 20 (where it is also HSRP Active), and MLS2 is configured as Root Bridge for VLAN 30 and 99

5.3.1 HQ Configuration

MLS1 is configured as primary root bridge for vlan 10,20,30,99,666
MLS2 is configured as primary root bridge for vlan 30,99

MLS1

```
MLS1(config)# spanning-tree vlan 10,20,30,666 root primary
MLS1(config)# spanning-tree vlan 30,99 root secundary
```

MLS2

```
MLS2(config)# spanning-tree vlan 30,99 root primary
MLS2(config)# spanning-tree vlan 10,20,666 root secondary
```

Note: The 'root primary' macro sets priority to the lowest multiple of 4096 below 24576. The 'root secondary' macro sets it to 28672. This assumes all other switches remain at default priority (32768).

PortFast is enabled on all access-layer ports connected to end devices:

Representative device: A-SW1 (same applies to A-SW2, A-SW3)

```
A-SW1(config)# interface range f0/1 - 2
A-SW1(config-if-range)# spanning-tree portfast
```

There are the verification commands and the output for representative devices:

```
MLS1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0060.7092.7110
             Cost        12
             Port        27(Port-channel1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0002.4A78.4598
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p
Po1              Root FWD 12        128.27   Shr
```

```
A-SW1#show spanning-tree interface f0/1 portfast
VLAN0001         enabled
VLAN0010         enabled
VLAN0020         disabled
VLAN0030         disabled
VLAN0099         disabled
VLAN0555         disabled
VLAN0666         disabled
```

5.3.2 Branch Office & Data Center Configuration

For both the Branch Office and the Data Center, the STP configuration is reduced to enabling PortFast globally on all access ports. Since these sites do not have redundant links or multiple switches, there is no need for Root Bridge designation — STP operates in its default state and PortFast is sufficient to eliminate unnecessary port initialization delays for end devices.

This configuration applies identically to SW4 (Branch Office) and R3 (Data Center):

```
SW4(config)# spanning-tree portfast default
```

here we have the verification command with the expected output

```
SW4#show spanning-tree interface f0/2 portfast
VLAN0001         enabled
VLAN0110         enabled
VLAN0120         enabled
VLAN0555         enabled
VLAN0666         enabled
```

5.4 IP address & IVR configuration

Two Inter-VLAN Routing methods are implemented: MLS SVI IVR for the Headquarter and Router-on-a-Stick for the Branch Office. HSRP is configured on MLS1 and MLS2 to provide Layer 3 first-hop redundancy with per-VLAN load balancing — MLS1 is Active for VLAN 10 and 20, MLS2 is Active for VLAN 30 and 99

5.4.1 HQ - IP addresses & SVI IVR

representative device - MLS1

```
MLS1(config)# ip routing

MLS1(config)# interface vlan 10
MLS1(config-vlan)# ip add 10.1.10.2 255.255.255.0
MLS1(config-vlan)# standby 10 ip 10.1.10.1
MLS1(config-vlan)# standby 10 priority 110
MLS1(config-vlan)# standby 10 preempt
MLS1(config-vlan)# no shut

MLS1(config-vlan)# interface vlan 20
MLS1(config-vlan)# ip add 10.1.20.2 255.255.255.0
MLS1(config-vlan)# standby 20 ip 10.1.20.1
MLS1(config-vlan)# standby 20 priority 110
MLS1(config-vlan)# standby 20 preempt
MLS1(config-vlan)# no shut

MLS1(config-vlan)# interface vlan 30
MLS1(config-vlan)# ip add 10.1.30.2 255.255.255.0
MLS1(config-vlan)# standby 30 ip 10.1.30.1
MLS1(config-vlan)# standby 30 priority 100
MLS1(config-vlan)# standby 30 preempt
MLS1(config-vlan)# no shut

MLS1(config-vlan)# interface vlan 99
MLS1(config-vlan)# ip add 10.1.99.2 255.255.255.240
MLS1(config-vlan)# standby 99 ip 10.1.99.1
MLS1(config-vlan)# standby 99 priority 100
MLS1(config-vlan)# standby 99 preempt
MLS1(config-vlan)# no shut
```

note: “ip routing” is used to enable routing capabilities on layer 3 switch

part of ‘show ip interfaces brief’ that include what we configured before

```
Vlan10                 10.1.10.2       YES manual up                    up
Vlan20                 10.1.20.2       YES manual up                    up
Vlan30                 10.1.30.2       YES manual up                    up
Vlan99                 10.1.99.2       YES manual up                    up
```

For MLS1 and MLS2 from Headquarter we have to configure some additional layer 3 feature, HSRP to provide first hop redundancy in case of one device going down.

Another detail that should be remarked is that MLS1 is configured as Active router for VLAN10 and VLAN20, and MLS2 is configured as Active router for VLAN 30 and VLAN99, both by setting priority to 110. This is a method of Load balancing traffic and is important for resource efficient use.

```
MLS1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           10.1.10.3       10.1.10.1
Vl20        20   110 P Active   local           10.1.20.3       10.1.20.1
Vl30        30   100 P Standby  10.1.30.3       local           10.1.30.1
Vl99        99   100 P Standby  10.1.99.3       local           10.1.99.1
```

Then we have to configure the link between distribution and core layer

This is distribution part of configuration

```
MLS1(config)# ip routing
MLS1(config)# interface f0/4
MLS1(config-if)# no switchport
MLS1(config-if)# ip add 10.1.0.2 255.255.255.252
MLS1(config-if)# no shut
```

This is core part of configuration

```
R1(config)# interface g0/0
R1(config-if)# ip add 10.1.0.1 255.255.255.252
R1(config-if)# no shut


R1(config-if)# interface g0/1
R1(config-if)# ip add 10.1.0.5 255.255.255.252
R1(config-if)# no shut
```

5.4.2 Branch Office - RoaS IVR

Due to its smaller scale, the Branch Office uses Router-on-a-Stick for Inter-VLAN Routing, as this approach does not require a dedicated Layer 3 switch and is well-suited for environments with a low number of VLANs.

Representative device: R2

```
R2(config)# interface g0/0.110
R2(config-subif)# encapsulation dot1Q 110
R2(config-subif)# ip add 10.2.110.1 255.255.255.0
R2(config-subif)# no shut

R2(config-subif)# interface g0/0.120
R2(config-subif)# encapsulation dot1Q 120
R2(config-subif)# ip add 10.2.120.1 255.255.255.0
R2(config-subif)# no shut

R2(config-subif)# interface g0/0.555
R2(config-subif)# encapsulation dot1Q 555 native
R2(config-subif)# ip address 10.2.255.1 255.255.255.0
R2(config-subif)# no shut

R2(config-subif)# interface g0/0
R2(config-subif)# no shut
```

Verification:

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.110 unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2     10.0.0.17       YES manual up                    up
GigabitEthernet0/0/0   10.0.0.10       YES manual up                    up
GigabitEthernet0/1/0   unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

5.4.3 Data Center - SVI IVR

Given that the Data Center infrastructure utilizes a single router equipped with an integrated 4-port EtherSwitch module, the implementation of Switch Virtual Interfaces (SVIs) is the most efficient method for Inter-VLAN Routing.

```
R3(config)# interface vlan 200
R3(config-vlan)# ip address 10.3.200.1 255.255.255.0
R3(config-vlan)# no shutdown

R3(config-vlan)# interface vlan 210
R3(config-vlan)# ip address 10.3.210.1 255.255.255.0
R3(config-vlan)# no shutdown

R3(config-vlan)# interface vlan 220
R3(config-vlan)# ip address 10.3.220.1 255.255.255.0
R3(config-vlan)# no shutdown
```

Verification done by using ‘ show ip interface brief ’

```
Vlan200                10.3.200.1      YES NVRAM  up                    up
Vlan210                10.3.210.1      YES NVRAM  up                    up
Vlan220                10.3.220.1      YES NVRAM  up                    up
```

5.4.4 WAN connections

Here we have assigned IP addresses to all WAN interfaces

```
R1(config)# interface g0/0/0
R1(config-if)# ip address 10.0.0.9 255.255.255.252
R1(config-if)# no shutdown

R1(config-if)# interface g0/2
R1(config-if)# ip address 10.0.0.13 255.255.255.252
R1(config-if)# no shutdown
```

```
R2(config)# interface g0/0/0
R2(config-if)# ip address 10.0.0.10 255.255.255.252
R2(config-if)# no shutdown

R2(config-if)# interface g0/2
R2(config-if)# ip address 10.0.0.17 255.255.255.252
R2(config-if)# no shutdown
```

```
R3(config)# interface f0/0
R3(config-if)# ip address 10.0.0.18 255.255.255.252
R3(config-if)# no shutdown

R3(config-if)# interface f0/1
R3(config-if)# ip address 10.0.0.14 255.255.255.252
R3(config-if)# no shutdown
```

```
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.0.0.18       YES NVRAM  up                    up
FastEthernet0/1        10.0.0.14       YES NVRAM  administratively down down
```

5.5 Routing configuration

Static routing is used throughout the topology to achieve full convergence. Each router and Layer 3 switch requires routes to reach all remote subnets.

5.5.1 Headquarter

First, default gateways should be configured at the distribution layer to enable end-devices within the headquarter to communicate with outside networks.

```
MLS1(config)# ip route 0.0.0.0 0.0.0.0 10.1.0.1
```

```
MLS2(config)# ip route 0.0.0.0 0.0.0.0 10.1.0.5
```

Also, traffic from outside arriving at core distribution should know how to get to end-devices within HQ, meaning we need to have a route for each VLAN.

For redundancy, R1 uses floating static routes (higher administrative distance) as backup paths toward HQ VLANs via MLS2 when MLS1 is unavailable.

```
R1(config)# ip route 10.1.10.0 255.255.255.0 10.1.0.2
R1(config)# ip route 10.1.10.0 255.255.255.0 10.1.0.6 10

R1(config)# ip route 10.1.20.0 255.255.255.0 10.1.0.2
R1(config)# ip route 10.1.20.0 255.255.255.0 10.1.0.6 10

R1(config)# ip route 10.1.30.0 255.255.255.0 10.1.0.6
R1(config)# ip route 10.1.30.0 255.255.255.0 10.1.0.2 10

R1(config)# ip route 10.1.99.0 255.255.255.0 10.1.0.6
R1(config)# ip route 10.1.99.0 255.255.255.0 10.1.0.2 10
```

Routing devices are attached to each network internally is the reason for BO and DC not needing routing configuration for internal traffic.

5.5.2 WAN

For WAN static routing configuration has been made. Even if there could be configured some sort of dynamic routing for better scalability, the purpose of static configuration is to demonstrate the true understanding of the concept. An advantage of static routes is stability.

Here are all static routes configuration for WAN interfaces

```
R1(config)# ip route 10.2.0.0 255.255.0.0 10.0.0.10
R1(config)# ip route 10.2.110.0 255.255.0.0 10.0.0.14 2

R1(config)# ip route 10.3.0.0 255.255.0.0 10.0.0.14
R1(config)# ip route 10.3.0.0 255.255.0.0 10.0.0.10 2
```

```
R2(config)# ip route 10.1.0.0 255.255.0.0 10.0.0.9
R2(config)# ip route 10.1.0.0 255.255.255.0 10.0.0.18 2

R2(config)# ip route 10.3.0.0 255.255.0.0 10.0.0.18
R2(config)# ip route 10.3.0.0 255.255.0.0 10.0.0.9 2
```

```
R3(config)# ip route 10.1.0.0 255.255.0.0 10.0.0.13
R3(config)# ip route 10.1.0.0 255.255.0.0 10.0.0.17 2

R3(config)# ip route 10.2.0.0 255.255.0.0 10.0.0.17
R3(config)# ip route 10.2.0.0 255.255.0.0 10.0.0.13 2
```

note: static floating routes are included as redundancy method in case of specific interfaces going down

5.6 Security Configuration

5.6.1 Ports-security

A–SW1,2,3

```
A-SW1(config)# interface range f0/1 - 2
A-SW1(config-if-range)# switchport port-security
A-SW1(config-if-range)# switchport port-security maximum 1
A-SW1(config-if-range)# switchport port-security mac-address sticky
A-SW1(config-if-range)# switchport port-security violation shutdown
```

```
A-SW1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)        (Count)
--------------------------------------------------------------------
        Fa0/1        1          0                0         Shutdown
        Fa0/2        1          0                0         Shutdown
--------------------------------------------------------------------
```

```
A-SW1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)        (Count)
--------------------------------------------------------------------
        Fa0/1        1          0                 0         Shutdown
        Fa0/2        1          0                 0         Shutdown
----------------------------------------------------------------------
A-SW1#show por
A-SW1#show port-security in
A-SW1#show port-security interface f0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0
```

5.6.2 DHCP snooping

A-SW1,2,3

```
A-SW1(config)# ip dhcp snooping
A-SW1(config)# ip dhcp snooping vlan 10,20,30,99

A-SW1(config)# in range f0/3 - 4
A-SW1(config-if)# ip dhcp snooping trust
A-SW1(config-if)# ip dhcp snooping limit rate 100

A-SW1(config-if)# in range f0/1 - 2
A-SW1(config-if-range)# ip dhcp snooping limit rate 10
```

```
A-SW1#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10,20,30,99
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                    Trusted    Rate limit (pps)
-----------------------      -------    ----------------
FastEthernet0/1              no         10
FastEthernet0/2              no         10
FastEthernet0/3              yes        100
FastEthernet0/4              yes        100
```

MLS1,2

```
MLS1(config)# ip dhcp snooping
MLS1(config)# ip dhcp snooping vlan 10,20,30,99


MLS1(config)# interface range f0/5 - 6
MLS1(config-if-range)# ip dhcp snooping trust
MLS1(config-if-range)# ip dhcp snooping limit rate 100

MLS1(config-if-range)# in range f0/1 - 2
MLS1(config-if-range)# ip dhcp snooping limit rate 100
```

```
MLS1#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10,20,30,99
DHCP snooping is operational on following VLANs:
none
Smartlog is configured on following VLANs:
none
Smartlog is operational on following VLANs:
none
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is enabled
   circuit-id default format: vlan-mod-port
   remote-id: 0002.4A78.4598 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

Interface                  Trusted    Allow option    Rate limit (pps)
-----------------------    -------    ------------    ----------------
FastEthernet0/1            no         no              100
  Custom circuit-ids:
FastEthernet0/2            no         no              100
  Custom circuit-ids:
FastEthernet0/5            yes        yes             100
  Custom circuit-ids:
FastEthernet0/6            yes        yes             100
  Custom circuit-ids:
```

5.6.3 BPDU guard

```
A-SW1(config)# spanning-tree portfast bpduguard default
```

```
A-SW1#show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is enabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is disabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0010                      1         0        0          3          4
VLAN0020                      1         0        0          1          2
VLAN0030                      1         0        0          1          2
VLAN0099                      1         0        0          1          2
VLAN0555                      1         0        0          1          2

---------------------- -------- --------- -------- ---------- ----------
7 vlans                       5         0        0          7         12
```

5.6.4 Dynamic Trunking Protocol (DTP)

To mitigate VLAN hooping and Switch spoofing in Site 1 (Headquarter) DTP must be disabled on all ports configured to add a layer of security to network architecture.

here A-SW3 is used as an example device for access layer:

```
A-SW3(config)# interface range f0/1 - 24, g0/1 - 2
A-SW3(config-if-range)# switchport nonegotiate
```

Expected output for and example interface (all others should be the same at the line highlighted)

```
A-SW3#show interfaces FastEthernet 0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (Sales)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
```

MLS1 as an exemple for distribution layer:

```
MLS1(config)# interface range f0/1 - 3
MLS1(config-if-range)# switchport nonegotiate

MLS1(config-if-range)# interface po1
MLS1(config-if)# switchport nonegotiate

MLS1(config-if)# interface range f0/7 - 24
MLS1(config-if-range)# switchport nonegotiate


MLS1(config-if-range)# interface range g0/1 - 2
MLS1(config-if-range)# switchport nonegotiate
```

command used for verification -> show interface f0/3 switchport

```
Name: Fa0/3
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 555 (Native-BH)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 10,20,30,99,555
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
```

5.7 Services Configuration

5.7.1 SSH configuration for remote management (MSL1, MLS2)

To remote access our layer 3 switch from site 1 (Headquarter) we use ssh for encrypted secure connections over the networks.

As for ssh configuration next configuration should be done prior:

1. Configure hostname (already configured in the beginning of documentation)
2. Configure domain
3. Create a user and a passwords used for authentication
4. Create RSA keys

5. Optional but recommended is to use SSH version 2 with at least 1024 bits key for better security
6. Implement only ssh method to be used as remote management (for security purposes)

```
MLS2(config)# ip domain-name lab.local
MLS2(config)# user admin privilege 15 secret admin
MLS2(config)# crypto key generate rsa
The name for the keys will be: MLS2.lab.local
Choose the size of the key modulus in the range of 360 to 2048 for
your
  General Purpose Keys. Choosing a key modulus greater than 512 may
take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

MLS2(config)# ip ssh version 2
MLS2(config-line)# line vty 0 15
MLS2(config-line)# transport input ssh
MLS2(config-line)# login local
```

verification command -> show ip ssh

```
MLS2#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication
retries: 3
```

For simplicity admin is used as an encrypted password, but in real environments a stronger password should be used. As for privilege, the admin was assigned the highest privileged 15 for total control over devices.
as shown above ssh version 2 was configured as it has better security

5.8 DHCPv4

DHCP is a service used to automatically assign IP addresses to devices in the network. In our example even though there aren't so many devices that need to be configured (for simplicity - this documentation is concentrated on demonstrating the understanding of concepts), in a real enterprise environment would be hundreds if not thousands of devices that need to be configured. For that reason DHCP is not only a time saver but using it become less likely to have a configuration error (which humans would tend to do configuring that many devices)

5.8.1 Site 1 - Headquarter

```
R1(config)# ip dhcp excluded-address 10.1.10.1 10.1.10.9
R1(config)# ip dhcp excluded-address 10.1.20.1 10.1.20.9
R1(config)# ip dhcp excluded-address 10.1.30.1 10.1.30.9


R1(config)# ip dhcp pool VLAN10
R1(dhcp-config)# network 10.1.10.0 255.255.255.0
R1(dhcp-config)# default-router 10.1.10.1
R1(dhcp-config)# dns-server 10.3.200.10

R1(dhcp-config)# ip dhcp pool VLAN20
R1(dhcp-config)# network 10.1.20.0 255.255.255.0
R1(dhcp-config)# default-router 10.1.20.1
R1(dhcp-config)# dns-server 10.3.200.10

R1(dhcp-config)# ip dhcp pool VLAN30
R1(dhcp-config)# network 10.1.30.0 255.255.255.0
R1(dhcp-config)# default-router 10.1.30.1
R1(dhcp-config)# dns-server 10.3.200.10
```

```
MLS1(config)# interface vlan 10
MLS1(config-if)# ip helper-address 10.1.0.1

MLS1(config-if)# interface vlan 20
MLS1(config-if)# ip helper-address 10.1.0.1

MLS1(config-if)# interface vlan 30
MLS1(config-if)# ip helper-address 10.1.0.1
```

```
MLS2(config)# interface vlan 10
MLS2(config-if)# ip helper-address 10.1.0.5

MLS2(config-if)# interface vlan 20
MLS2(config-if)# ip helper-address 10.1.0.5

MLS2(config-if)# interface vlan 30
MLS2(config-if)# ip helper-address 10.1.0.5
```

verification command -> show ip dhcp pool

Here VLAN10 pool stats can be seen

```
R1#show ip dhcp pool

Pool VLAN10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.1.10.1            10.1.10.1        - 10.1.10.254       0    / 3     / 254
```

As for the distribution layer, in the imagine below “show ip interface vlan 10” command is used to verify if the helper address was configured.

```
MLS1#show ip interface vlan 10
Vlan10 is up, line protocol is up
  Internet address is 10.1.10.2/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is 10.1.0.1
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Null turbo vector
  IP multicast fast switching is disabled
```

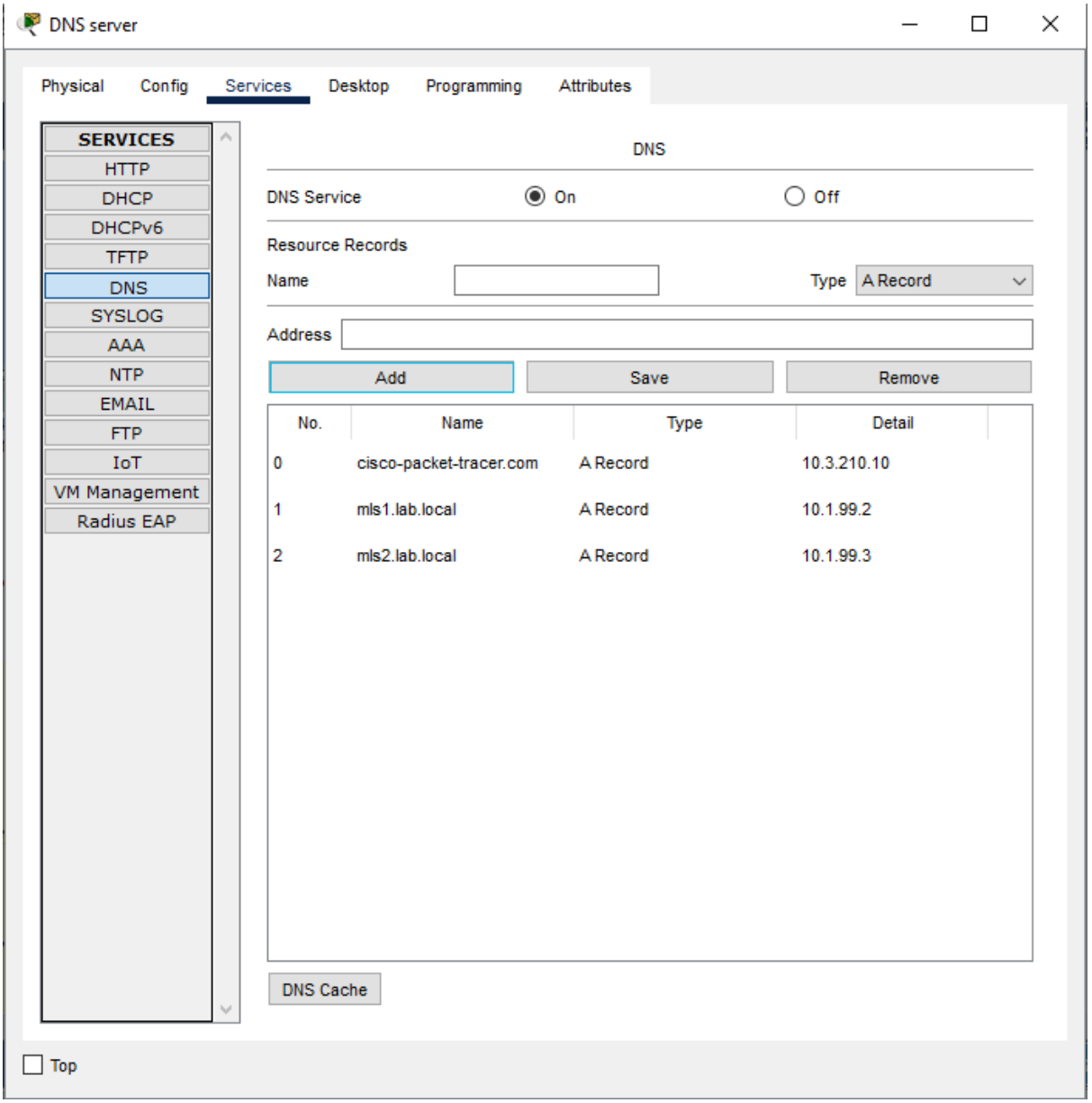
5.9 DNS Server config

First on R1 must be used

```
R1(config)# ip name-server 10.3.200.10
```

in order to enable R1 to locate the dns server which resolves domain queries.

Below are domains assigned to management ip addresses of MLS1 and MLS2 from Headquarter to enable ssh connection through their name and location.



To verify if DNS is correctly configured we should try ‘nslookup mls1.lab.local’ command in the command prompt for example on PC0.

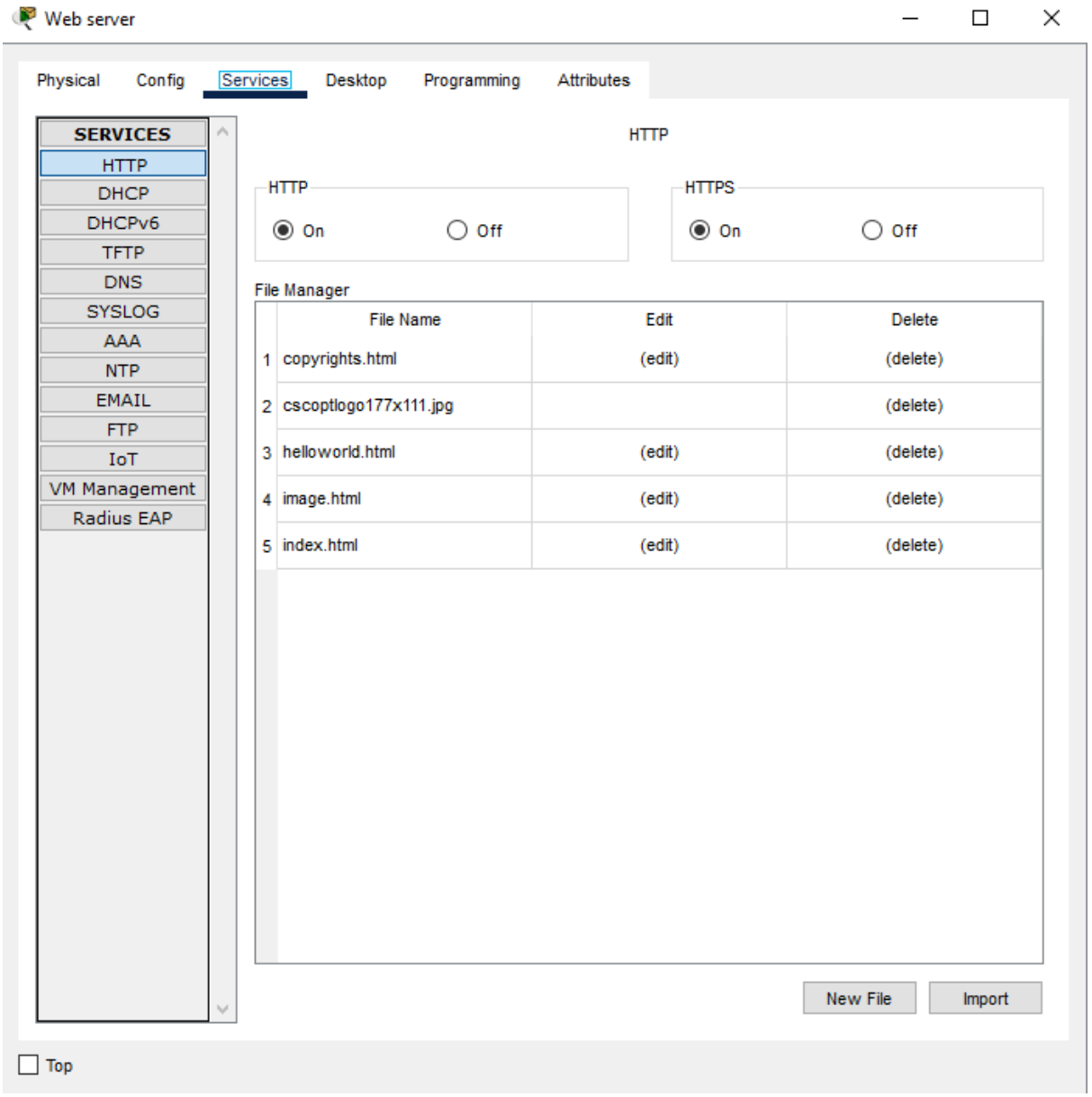
```
C:\>nslookup mls1.lab.local

Server: [10.3.200.10]
Address:  10.3.200.10

Non-authoritative answer:
Name:   mls1.lab.local
Address:   10.1.99.2

C:\>nslookup mls2.lab.local

Server: [10.3.200.10]
Address:  10.3.200.10

Non-authoritative answer:
Name:   mls2.lab.local
Address:   10.1.99.3
```

Here we can see the DNS server ip address and the ip address corresponding to the name we use as input in nslookup command.

5.10 HTTP/S configuration

To demonstrate application-layer service deployment, the Web Server (10.3.210.10) in the Data Center was configured to host a simple web page accessible via HTTP and HTTPS.

Configuration on Web Server:
In Cisco Packet Tracer, the HTTP and HTTPS services were enabled by turning them on from the Services tab of the server. A basic HTML page was used as the default index.



Verification:
From a client PC in the Headquarter (PC1, VLAN 10), the web browser was opened and the following URLs were entered:

- `http://10.3.210.10` – The webpage loaded successfully, confirming HTTP connectivity.
- `https://10.3.210.10` – The HTTPS connection was established

Web Browser X

< > URL http://10.3.210.10 Go Stop

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6. Testing and Verification

This section validates the network against the design objectives using a structured approach: basic reachability, high availability mechanisms, essential network services, and security enforcement.

6.1 Functional Connectivity Tests

First, we ensure that all devices can communicate as intended.

6.1.1 Intra-VLAN Communication

Test within HQ, VLAN 30, PC5 to PC6:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.30.11

Pinging 10.1.30.11 with 32 bytes of data:

Reply from 10.1.30.11: bytes=32 time<1ms TTL=128
Reply from 10.1.30.11: bytes=32 time<1ms TTL=128
Reply from 10.1.30.11: bytes=32 time<1ms TTL=128
Reply from 10.1.30.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Both PCs obtained IPs via DHCP and can communicate directly through the access and distribution switches.

6.1.2 Inter-VLAN & Inter-Site Communication

From PC4 (10.1.20.11) to PC7 (10.2.110.10):

```
C:\>ping 10.2.110.10

Pinging 10.2.110.10 with 32 bytes of data:

Reply from 10.2.110.10: bytes=32 time<1ms TTL=125
Reply from 10.2.110.10: bytes=32 time<1ms TTL=125
Reply from 10.2.110.10: bytes=32 time<1ms TTL=125
Reply from 10.2.110.10: bytes=32 time=9ms TTL=125

Ping statistics for 10.2.110.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms
```

From PC3 (HQ, 10.1.20.11) to DNS Server (DC, 10.3.200.10):

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.3.200.10

Pinging 10.3.200.10 with 32 bytes of data:

Request timed out.
Reply from 10.3.200.10: bytes=32 time<1ms TTL=125
Reply from 10.3.200.10: bytes=32 time<1ms TTL=125
Reply from 10.3.200.10: bytes=32 time<1ms TTL=125

Ping statistics for 10.3.200.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

All inter-site and inter-VLAN pings succeed, proving correct routing and trunk configurations.

6.1.3 WAN Link Reachability

Verify point-to-point links between routers:

```
R1#ping 10.0.0.10 (R2)

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R1#ping 10.0.0.14 (R3)

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.14, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

```
R2#ping 10.0.0.18 (R3)

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.18, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

All WAN interfaces are up and responding.

6.2 High Availability & Redundancy Tests

6.2.1 HSRP Failover

A continuous ping was run from PC3 (10.1.20.10) to the default gateway VIP (10.1.10.1). During the ping, all interfaces were administratively down on the Active router MLS1:

```
C:\>ping -t 10.1.20.1

Pinging 10.1.20.1 with 32 bytes of data:


Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
Reply from 10.1.20.1: bytes=32 time<1ms TTL=255
```

some pings was lost; MLS2 (Standby) promoted to Active:

6.3 Network Services Verification

6.3.1 DHCP Address Assignment

On PC3, the IP parameters were checked:

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.9615.9AC4
   Link-local IPv6 Address.........: FE80::201:96FF:FE15:9AC4
   IPv6 Address....................: ::
   IPv4 Address....................: 10.1.20.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     10.1.20.1
   DHCP Servers....................: 10.1.0.1
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-B8-27-20-25-00-01-96-15-9A-C4
   DNS Servers.....................: ::
                                     10.3.200.10
```

The address comes from the correct pool, and the DNS suffix is properly assigned.

6.3.2 DNS Name Resolution

From the PC, the hostnames of the management switches were resolved: (as already shown in 5.9 section)

```
C:\>nslookup mls1.lab.local

Server: [10.3.200.10]
Address:  10.3.200.10

Non-authoritative answer:
Name:   mls1.lab.local
Address:   10.1.99.2

C:\>nslookup mls2.lab.local

Server: [10.3.200.10]
Address:  10.3.200.10

Non-authoritative answer:
Name:   mls2.lab.local
Address:   10.1.99.3
```

DNS is operational and improves network manageability.

6.4 Security Feature Validation

6.4.1 Port Security

A device with an unknown MAC address was connected to A-SW1 port Fa0/1. The port immediately went into err-disabled state:

```
A-SW1#show port-security interface f0/1
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 000A.F3E4.BDC5:10
Security Violation Count   : 1
```

This prevents unauthorized devices from accessing the network.

6.4.3 BPDU Guard & DTP

Connecting an unauthorized switch to an PortFast-enabled access port triggers BPDU guard, which places the port in `err-disabled` state. The `switchport nonegotiate` command was verified on access ports with `show interfaces switchport`, showing "Negotiation of Trunking: Off". These measures mitigate STP manipulation and VLAN hopping attacks.

7. Conclusions

This project successfully demonstrated the end-to-end design, configuration, and verification of a multi-site enterprise network using Cisco technologies. The network was built with a strong emphasis on high availability, employing HSRP for Layer 3 gateway redundancy, EtherChannel for link resilience, and Rapid PVST+ for loop-free Layer 2 topology with per-VLAN load balancing.

A structured IPv4 addressing plan using VLSM ensured efficient address allocation while allowing for future scalability. VLAN segmentation and Inter-VLAN routing via both SVI and Router-on-a-Stick showcased flexible implementation according to site size. Security was integrated from the ground up with Port Security, DHCP Snooping, BPDU Guard, and DTP negotiation disabled, protecting the network against common Layer 2 attacks.

Core network services—DHCP, DNS, HTTP/HTTPS, and SSH—were deployed and tested successfully. The comprehensive testing phase confirmed end-to-end connectivity, HSRP failover, and proper operation of all security mechanisms.

A routing issue encountered during configuration (R3 interface administratively down) provided practical troubleshooting experience and underscored the importance of verifying physical and data-link layer states before debugging upper-layer problems.

Moving forward, this infrastructure can be extended with dynamic routing protocols like OSPF, IPv6 addressing, wireless LAN integration, and VPN remote access. This project reflects the skill set required for a junior network administrator or NOC engineer role and serves as a solid foundation for continued professional growth.